

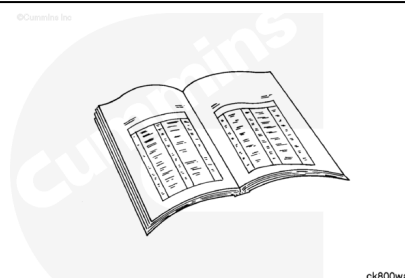
## Trainings ▾

### Preparatory Steps

#### with Mechanically Actuated Injector

##### **⚠ WARNING ⚠**

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**



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##### **⚠ WARNING ⚠**

**Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.**

##### **⚠ CAUTION ⚠**

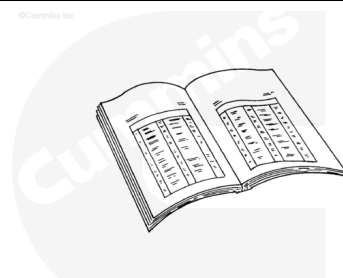
**Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be disposed of in accordance with local environmental regulations.**

- Disconnect the battery cables. Refer to Procedure 013-009 (Battery Cables and Connections) in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Remove the fuel filters. Refer to Procedure 006-015 (Fuel Filter (Spin-On Type)) in Section 6. (</qs3/pubsys2/xml/en/procedures/20/20-006-015-tr.html>)
- Disconnect the fuel supply line from the fuel filter head assembly. Refer to Procedure 006-024 (Fuel Supply Lines) in Section 6. (</qs3/pubsys2/xml/en/procedures/20/20-006-024-tr.html>)

## with Electronically Actuated Injector

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- Remove the fuel filters. Refer to Procedure 006-015 (Fuel Filter (Spin-On Type)) in Section 6. (</qs3/pubsys2/xml/en/procedures/20/20-006-015-tr.html>)
- Remove the fuel supply lines from the fuel filter head assembly. Refer to Procedure 006-024 (Fuel Supply Lines) in Section 6. (</qs3/pubsys2/xml/en/procedures/20/20-006-024-tr.html>)
- Disconnect the fuel pressure sensor. Refer to Procedure 019-398 (Fuel Supply Pump Pressure Sensor) in the Troubleshooting and Repair Manual, Electronic Control System QSK19 CM850 Modular Common Rail System, Bulletin 4021493. (</qs3/pubsys2/xml/en/procedures/13/13-019-398.html>)
- Disconnect the fuel temperature sensor. Refer to Procedure 019-119 (Engine Coolant Temperature Sensor)

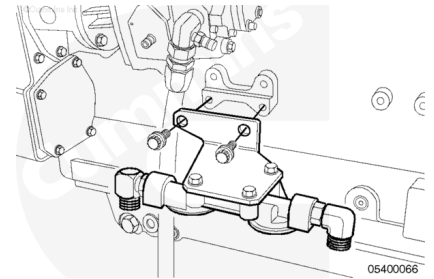
in the Troubleshooting and Repair Manual, Electronic Control System QSK19 CM850 Modular Common Rail System, Bulletin 4021493.  
(/qs3/pubsys2/xml/en/procedures/13/13-019-119.html)

## Remove

### with Mechanically Actuated Injector

Remove the two mounting capscrews from the fuel filter head bracket.

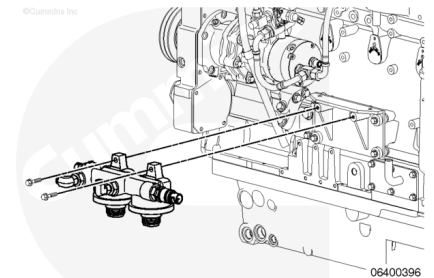
Remove the fuel filter head and bracket.



### with Electronically Actuated Injector

Remove the two capscrews that attach the fuel filter head assembly to the vibration isolator plate.

Remove the fuel filter head assembly from the vibration isolator plate.

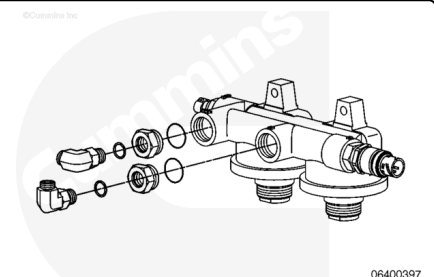


## Disassemble

### with Electronically Actuated Injector

Remove the union fittings and adapters from the fuel filter head assembly.

Remove and discard the o-rings from the fittings.

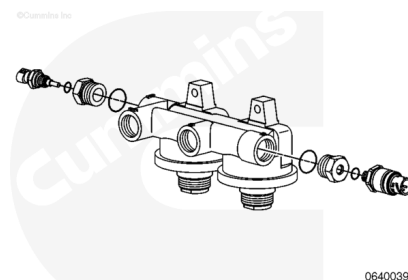


Remove the fuel pressure sensor. Refer to Procedure 019-398 (Fuel Supply Pump Pressure Sensor) in the Troubleshooting and Repair Manual, Electronic Control System QSK19 CM850 Modular Common Rail System, Bulletin 4021493. (</qs3/pubsys2/xml/en/procedures/13/13-019-398.html>)

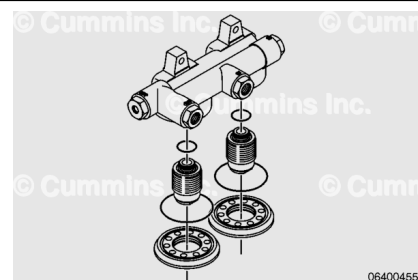
Remove the fuel temperature sensor. Refer to Procedure 019-119 (Engine Coolant Temperature Sensor) in the Troubleshooting and Repair Manual, Electronic Control System, QSK19 CM850, Modular Common Rail System, Bulletin 4021493. (</qs3/pubsys2/xml/en/procedures/13/13-019-119.html>)

Remove the sensor adapters.

Discard the o-rings.



Remove filter head adapter spud and adapter ring. Remove and discard the 4 o-rings.



## Clean and Inspect for Reuse

with Electronically Actuated Injector

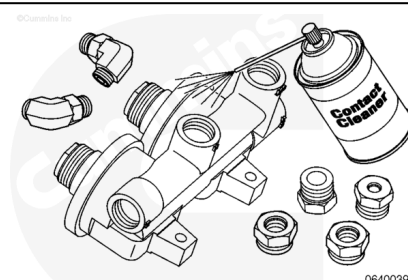
### **⚠ WARNING ⚠**

**Some solvents are flammable and toxic. Read the manufacturer's instructions before using.**

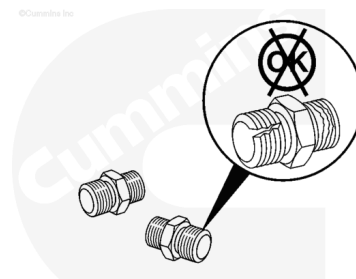
### **⚠ WARNING ⚠**

**When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.**

Clean all parts with contact cleaner, Part Number 3824510, or equivalent.

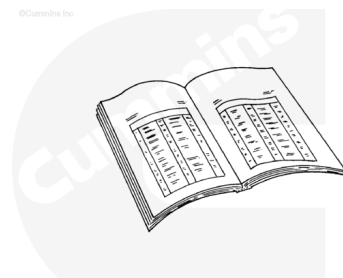


Inspect the union fittings for stripped or damaged threads, cracks, and other damage. Replace any damaged union fittings.



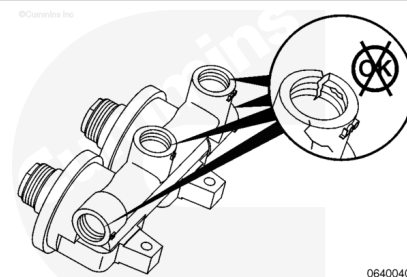
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Inspect the fuel pressure and fuel temperature sensor. Refer to Procedure 019-398 (Fuel Supply Pump Pressure Sensor) in the Troubleshooting and Repair Manual, Electronic Control System QSK19 CM850 Modular Common Rail System, Bulletin 4021493. ([/qs3/pubsys2/xml/en/procedures/13/13-019-398.html](http://qs3/pubsys2/xml/en/procedures/13/13-019-398.html)) and Refer to Procedure 019-119 (Engine Coolant Temperature Sensor) in the Troubleshooting and Repair Manual, Electronic Control System, QSK19 CM850, Modular Common Rail System, Bulletin 4021493. ([/qs3/pubsys2/xml/en/procedures/13/13-019-119.html](http://qs3/pubsys2/xml/en/procedures/13/13-019-119.html))



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Inspect the fuel filter head assembly for corrosion, cracks, and stripped or damaged threads. Damaged threads that attach the fuel filters or union fittings can **not** be repaired.



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## Assemble

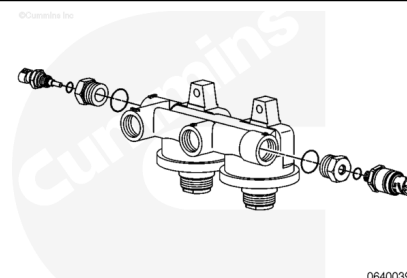
### with Electronically Actuated Injector

Install the new o-rings on the sensor adapters and install them into the ends of the filter head.

**Torque Value:** 156 n·m [115 ft-lb]

Install the fuel pressure sensor and the fuel temperature sensor. Refer to Procedure 019-398 (Fuel Supply Pump Pressure Sensor) in the Troubleshooting and Repair Manual, Electronic Control System QSK19 CM850 Modular Common Rail System, Bulletin 4021493.

([/qs3/pubsys2/xml/en/procedures/13/13-019-398.html](http://qs3/pubsys2/xml/en/procedures/13/13-019-398.html)) and



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Refer to Procedure 019-119 (Engine Coolant Temperature Sensor) in the Troubleshooting and Repair Manual, Electronic Control System, QSK19 CM850, Modular Common Rail System, Bulletin 4021493.  
(/qs3/pubsys2/xml/en/procedures/13/13-019-119.html).

Install new o-rings on the union fitting and install the inlet and outlet fittings in the filter head.

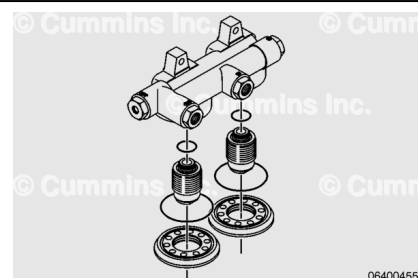
**Torque Value:** 62 n•m [ 46 ft-lb ]

Install new o-rings above the filter head adapter spuds. Apply thread locker to the smaller threaded portion of the spud. Install the spuds in the filter head.

**Torque Value:** 87 n•m [ 65 ft-lb ]

Install new o-rings above filter head adapter rings. Apply thread locking compound to inside of the threads. Thread the rings onto the spuds until the o-ring contacts the filter head. Use care to make sure the o-ring is properly seated.

**Torque Value:** 87 n•m [ 65 ft-lb ]



## Install

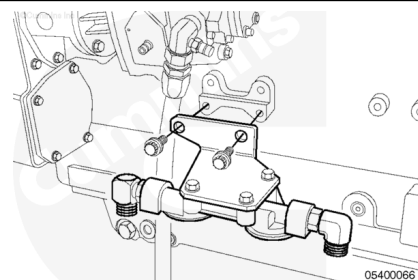
### with Mechanically Actuated Injector

Install the fuel filter head bracket and fuel filter head assembly.

Install the two mounting capscrews.

Tighten the capscrews.

**Torque Value:** 55 n•m [ 40 ft-lb ]

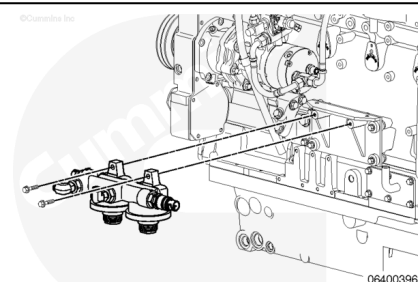


### with Electronically Actuated Injector

Install the fuel filter head assembly and two mounting capscrews to the vibration isolator mounting plate.

Tighten the capscrews.

**Torque Value:** 55 n•m [ 40 ft-lb ]

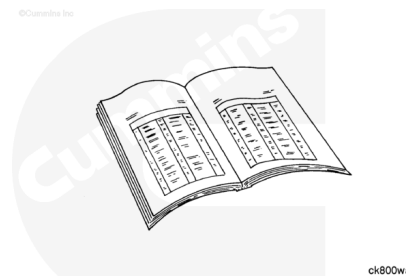


## Finishing Steps

### with Mechanically Actuated Injector

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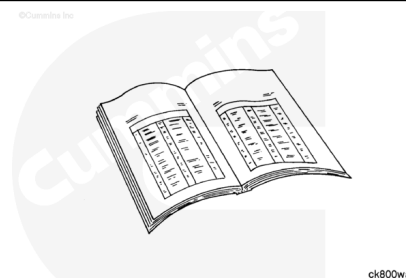
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- Connect the fuel supply lines to the fuel filter head assembly. Refer to Procedure 006-024 (Fuel Supply Lines) in Section 6. (</qs3/pubsys2/xml/en/procedures/28/28-006-024-tr.html>)
- Install the fuel filters. Refer to Procedure 006-015 (Fuel Filter (Spin-On Type)) in Section 6. (</qs3/pubsys2/xml/en/procedures/20/20-006-015-tr.html>)
- Connect the battery cables. Refer to Procedure 013-009 (Battery Cables and Connections) in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Operate the engine and check for leaks.

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- Connect the fuel temperature sensor. Refer to Procedure 019-119 (Engine Coolant Temperature Sensor) in the Troubleshooting and Repair Manual, Electronic Control System QSK19 CM850 Modular Common Rail System, Bulletin 4021493. (</qs3/pubsys2/xml/en/procedures/13/13-019-119.html>)

- Connect the fuel pressure sensor. Refer to Procedure 019-398 (Fuel Supply Pump Pressure Sensor) in the Troubleshooting and Repair Manual, Electronic Control System QSK19 CM850 Modular Common Rail System, Bulletin 4021493.  
(/qs3/pubsys2/xml/en/procedures/13/13-019-398.html)
- Install the fuel supply lines on the fuel filter head assembly. Refer to Procedure 006-024 (Fuel Supply Lines) in Section 6. (/qs3/pubsys2/xml/en/procedures/20/20-006-024-tr.html)
- Install the fuel filters. Refer to Procedure 006-015 (Fuel Filter (Spin-On Type)) in Section 6.  
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- Connect the battery cables. Refer to Procedure 013-009 (Battery Cables and Connections) in Section 13.  
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- Operate the engine and check for leaks.